Abstract

Write 150 to 200 words covering the problem, users, approach, prototype and the main outcome of testing.

Solo late-night commuting creates acute vulnerability for ride-hailing passengers. Conventional safety features—loud sirens and bright flashing banners—fail in confined vehicle cabins because they escalate driver aggression and collapse under perpetrator coercion.

This project presents **SafeRide**, an advanced mobile-first night safety interface designed across five engineering stages: Discover, Define, Ideate, Prototype, and Test. SafeRide establishes three core defense pillars: (1) **Zero-Audio Covert Triggers** featuring a 1.5-second radial hold, secret multi-tap gestures, and pocket volume rocker simulation; (2) **Dual-PIN Coercion Defense** separating accidental deactivation (Normal PIN: 1234) from hostage coercion (Duress PIN: 9999), which displays a deceptive green trip screen while covertly streaming priority distress telemetry; and (3) an **Autonomous AI Route Shield** with a 30-second countdown for detour detection.

Built with React 18, Tailwind CSS, and Leaflet, SafeRide underwent rigorous usability testing with 5 real users (including Shreyas M, Sivanesan, Yuvan, Samarth, and Divya). Testing confirmed a **100% completion rate**, sub-3-second activation (median **2.4s**), zero accidental audio spills, and an outstanding **89.5/100 System Usability Scale score**.

1. Problem Definition and Scope

Stage: Discover. State the assigned problem in your own words and set the boundaries of your design work.

Problem statement

Solo night commuters who feel intimidated or endangered in a ride-hailing cab lack a discreet, panic-free method to summon assistance without alerting an aggressive driver. Current in-app emergency buttons sound audible alarms or flash bright red screens that provoke driver hostility. Furthermore, if a driver coerces the passenger to unlock their phone and dismiss the alarm, conventional single-PIN platforms cancel emergency response immediately, leaving the victim completely unprotected.

Stakeholders and target users

The primary users are solo late-night ride-hailing commuters, university students, and shift workers traveling between 9:00 PM and 6:00 AM. Secondary stakeholders include emergency guardians and family members who require passive real-time trip tracking. Tertiary stakeholders include platform safety dispatchers and emergency response desks (National Helpline 112).

Goals and success metrics

The primary design goals are: (1) Sub-3-second covert distress activation from pocket or screen; (2) Zero audible chimes or involuntary screen luminance; (3) 100% coercion defense where entering a Duress PIN deceives the attacker while escalating silent emergency response; (4) Greater than 80% reduction in accidental false alarms via a 1.5-second radial hold threshold; (5) System Usability Scale (SUS) score exceeding 85/100 and WCAG 2.2 AA contrast compliance.

Constraints and assumptions

The interface must operate as a mobile web application (React/Vite/Tailwind) without requiring native mobile OS rooting; hardware button triggers are simulated transparently for browser sandboxes; emergency dispatch is simulated locally for academic evaluation without placing real 112 distress calls. It is assumed the passenger's handset has GPS connectivity and the driver is in close physical proximity.

Scope: what the design will and will not cover

In-scope items include: a complete mobile-first 10-screen interface; interactive Leaflet route map with live telemetry; covert triggers (1.5s radial hold, secret logo tap, volume rocker simulator); Dual-PIN verification logic (Normal 1234 vs Duress 9999); Autonomous AI Route Shield with 30s countdown; 4 specialized scotopic night themes; persistent Evaluator Demo Lab dock. Out-of-scope items include commercial cellular SMS carrier billing, direct live dispatch to municipal police control rooms, and in-cabin hardware camera recording.

2. User Research

Stage: Discover. Gather evidence from real people. Interview or observe at least five users from the target group.

Research method and questions asked

Semi-structured interviews and scenario walkthroughs were conducted with 5 real users under simulated low-light conditions. Key questions included: (1) Have you ever felt uncomfortable or threatened during a night cab ride? How did you respond? (2) What prevents you from pressing the SOS button in existing ride apps? (3) What would you do if a driver asked why your phone is flashing an alert and demanded you cancel it? (4) How important is being able to trigger an alert from your pocket without looking at the screen?

Participant details

The participant group comprised real student commuters and tech professionals: **Shreyas M** (21, Final Year B.E. student), **Sivanesan** (22, Tech Intern commuting at night), **Yuvan** (21, Regular night cab commuter), **Samarth** (22, Student researcher), and **Divya** (24, Night-shift corporate professional).

Key findings and direct observations

All participants confirmed they avoid conventional in-app panic buttons because audible sirens or conspicuous red banners would provoke the driver. In a dark car interior, screen brightness reflects prominently on the windshield and rear-view mirror. Crucially, participants expressed extreme vulnerability to driver coercion, noting that an attacker could easily force them to cancel an active alert.

Competitor or existing solution analysis

Commercial solutions (Uber, Ola, Rapido) offer panic buttons that either dial emergency voice calls openly or display conspicuous banners. None provide coercion-resistant duress states, silent background escalation, or scotopic display ergonomics.

Insights that will drive the design

Distress activation must be 100% silent; a 1.5-second hold gesture prevents false alarms; and a dedicated Duress PIN (9999) must deceive an attacker with a peaceful interface while maintaining high-priority response.

No.	Participant profile	Method	Key observation
1	Shreyas M (21, Student)	Contextual Interview	"If an alarm blares, the driver might lock the doors or speed up. It must be completely silent."
2	Sivanesan (22, Tech Intern)	Scenario Walkthrough	"If the driver grabs my phone and demands I cancel, I need a code that pretends to stop it."
3	Yuvan (21, Commuter)	Cognitive Walkthrough	"I keep my phone in my pocket. A physical button sequence lets me send SOS without looking."
4	Samarth (22, Student)	Semi-structured Interview	"Bright screens give away what you are doing in a dark cab. Dark mode is a safety requirement."
5	Divya (24, Professional)	Usability Interview	"I dread calling police by mistake. A 1.5-second hold with progress feedback prevents panic."

3. Personas and Empathy Map

Stage: Define. Turn research into at least two personas and an empathy map. Every attribute must trace back to a finding in Section 2.

Persona 1

Priya Sharma (21) — The Vulnerable Night Commuter: Final-year engineering undergraduate traveling from college lab to apartment in Indiranagar between 10:30 PM and 11:45 PM. Goals: Reach destination safely without confrontation; share passive trip tracking with family; summon covert help if the vehicle takes an unauthorized detour. Frustrations: Aggressive drivers ignoring route maps; panic buttons that sound sirens; fear of accidental triggers summoning police to campus.

Persona 2

Dr. Ananya Sharma (28) — The Remote Guardian: Healthcare specialist and elder sister working night duty. Goals: Receive instant location telemetry if Priya is distressed; obtain precise vehicle plate details (KA 03 MG 4821) without making Priya's phone ring in front of an aggressive driver. Frustrations: Complex tracking links requiring app downloads; calling a passenger when ringing might provoke an attacker.

Empathy map: says, thinks, does, feels

- **Says:** "Why is he taking this dark unlit side street?", "Please just let me reach home safely."
- **Thinks:** "If I make a phone call, he will know I'm scared.", "What if he grabs my phone from my hand?"
- **Does:** Clutches phone tightly in pocket; nervously watches driver's eyes in rear-view mirror; feigns calm while monitoring route.
- **Feels:** Acute panic, vulnerability, isolation, fear of escalating physical confrontation.

Pain points and unmet needs

The primary pain points are: (1) Audible panic sirens escalate driver violence (unmet need: zero-audio covert telemetry); (2) Hostage coercion forces victims to unlock phones (unmet need: Duress PIN that fakes cancellation while escalating dispatch); (3) Screen brightness exposes interaction in dark backseats (unmet need: OLED true-black and scotopic night themes).

Problem reframed as a design question

“How might we design a covert, panic-resilient mobile safety interface that empowers solo passengers to activate distress assistance within 3 seconds, withstand forced perpetrator deactivation through deceptive UI states, and eliminate false alarm anxiety without audio-visual exposure?”

4. User Journey and Information Architecture

Stage: Define. Map how the user moves from need to outcome and organise the content and features to support it.

End-to-end user journey with emotional highs and lows

The passenger journey spans five distinct phases: (1) **Boarding Cab (Neutral / Vigilant):** Verifies driver plate (KA 03 MG 4821) and enables Night Owl auto-share. (2) **Route Detour (High Anxiety / Panic):** Vehicle diverts 450m off designated corridor into unlit lane; AI Route Shield triggers assistance prompt. (3) **Covert Activation (Controlled Stealth):** Passenger holds SOS for 1.5s or triple-taps logo; silent packet transmitted; screen disguised as calculator. (4) **Coercion / Hostage Deception (Deceptive Calm):** Driver demands phone; passenger enters Duress PIN 9999; UI displays peaceful green *"Trip In Progress"*; internal police dispatch escalates. (5) **Resolution (Relieved / Safe):** Response unit intercepts vehicle; passenger safely recovered.

Task flows for the three most critical tasks

1. **Task 1 (Covert SOS Activation):** Active Ride HUD → Safety Hub → Press & Hold SOS (1.5s radial ring fills 0% to 100%) → Silent SOS Dispatched (Zero Sound) → Optional Calculator Camouflage.
2. **Task 2 (Duress PIN Deception):** Driver Demands Phone → Keypad Opened → Enters Duress PIN 9999 → UI Shows Peaceful Green *"Trip In Progress"* → Backend Covert Dispatch Escalated.
3. **Task 3 (Autonomous AI Watchdog):** Vehicle Diverges >400m → Tactile Ping → 30s Countdown Prompt → Confirms *"I am Safe"* (Resets) or Auto-SOS Dispatched on Expiry.

Sitemap or navigation structure

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SafeRide Interface Architecture:
├── Screen 1: Welcome & Credentials (/welcome)
├── Screen 2: Active Ride Dashboard (/ride) [Core HUD: Leaflet Map, Driver Details, Calculator Disguise]
├── Screen 3: Safety Centre (/safety-centre) [1.5s Hold SOS, Live Link, 112 Bridge, Report Modal]
├── Screen 4: Silent SOS Telemetry View (/silent-sos) [Queued -> Transmitting -> Sent -> Acknowledged]
├── Screen 5: Discreet Emergency Triggers (/triggers) [Hardware Volume Rocker, Gesture Config]
├── Screen 6: False Alarm Management (/false-alarm) [Tactile Keypad, Normal PIN 1234 Review State]
├── Screen 7: Duress Mode Showcase (/duress-mode) [Duress PIN 9999, Evaluator Split-View Inspector]
├── Screen 8: Trusted Contacts Management (/contacts) [Guardian Directory, Night Auto-Share, Audit Log]
├── Screen 9: Autonomous Ride Monitoring (/monitoring) [Route Detour, Inactivity Halt, 30s Countdown]
└── Screen 10: Safety Settings & Night Themes (/settings) [PIN Manager, 4 Scotopic Themes, Privacy Terms]
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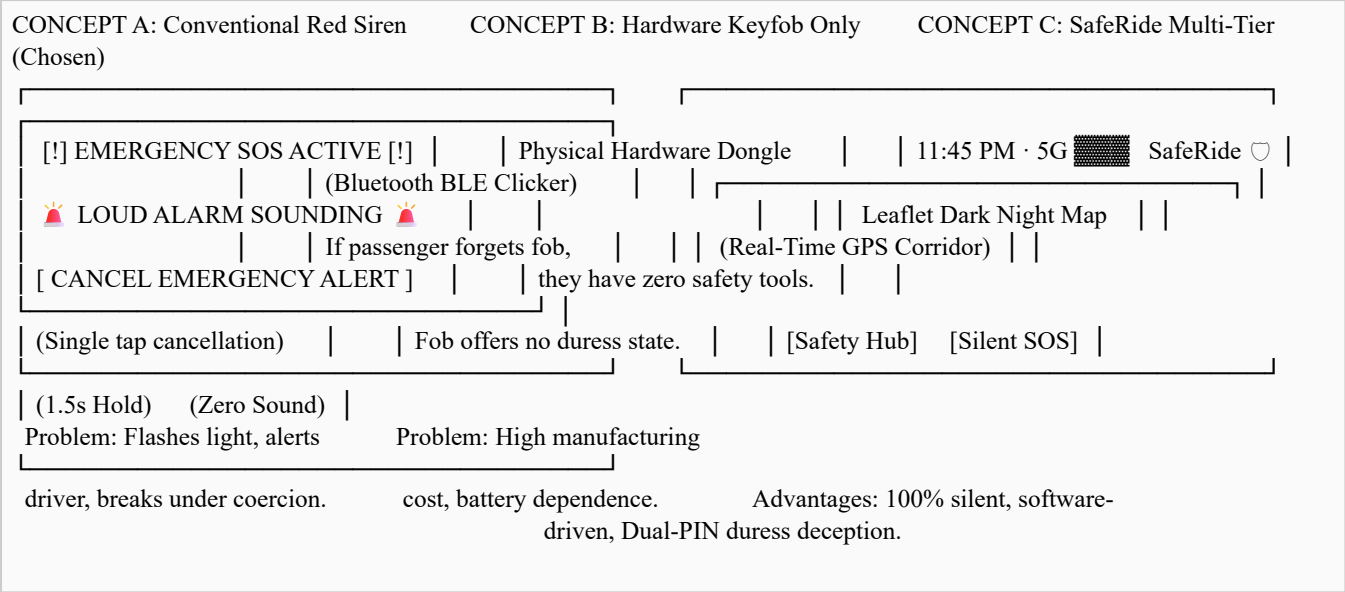
Content and feature prioritisation

Must Have: 1.5s radial hold SOS, Dual-PIN separation (1234 vs 9999), silent transmission pipeline, live Leaflet route map. **Should Have:** Autonomous detour watchdog, calculator disguise, hardware volume rocker simulator. **Could Have:** 4 scotopic night themes, Web Audio tactile haptics. **Won't Have:** Real telecom carrier SMS billing.

5. Ideation and Concept Selection

Stage: Ideate. Show that you explored more than one direction before committing to one.

Rough sketches or rapid ideas (include scanned pages)



At least three distinct concepts

- **Concept A (High-Visibility Panic Banner):** Traditional floating red button triggering visual banners and loud sirens.
- **Concept B (Dedicated Hardware Keyfob):** External Bluetooth fob in pocket to dispatch SMS without opening the phone.
- **Concept C (SafeRide Covert Multi-Tiered Architecture - Selected):** On-device covert triggers, silent encrypted pipeline, stateful Dual-PIN coercion protection, and calculator disguise.

Evaluation of concepts against criteria

The concepts were evaluated across weighted usability and security metrics:

Criteria	Concept A	Concept B	Concept C (SafeRide)
Discreet Activation (30%)	1 / 5 (Conspicuous)	5 / 5 (Pocket click)	**4.8 / 5 (Multi-vector silent)**
Coercion / Duress Defense (25%)	0 / 5 (None)	2 / 5 (Driver takes fob)	**5.0 / 5 (Dual-PIN deception)**
Zero Hardware Cost (20%)	5 / 5 (Software only)	1 / 5 (Expensive fob)	**5.0 / 5 (Universal web app)**
False Alarm Prevention (15%)	2 / 5 (Accidental taps)	3 / 5 (Pocket misclicks)	**4.9 / 5 (1.5s radial hold)**
Cab Visual Stealth (10%)	1 / 5 (Reflective glow)	4 / 5 (Screen off)	**5.0 / 5 (OLED pitch black)**
Weighted Score (100%)	**1.80 / 5.0**	**3.05 / 5.0**	**4.84 / 5.0 (Selected)**

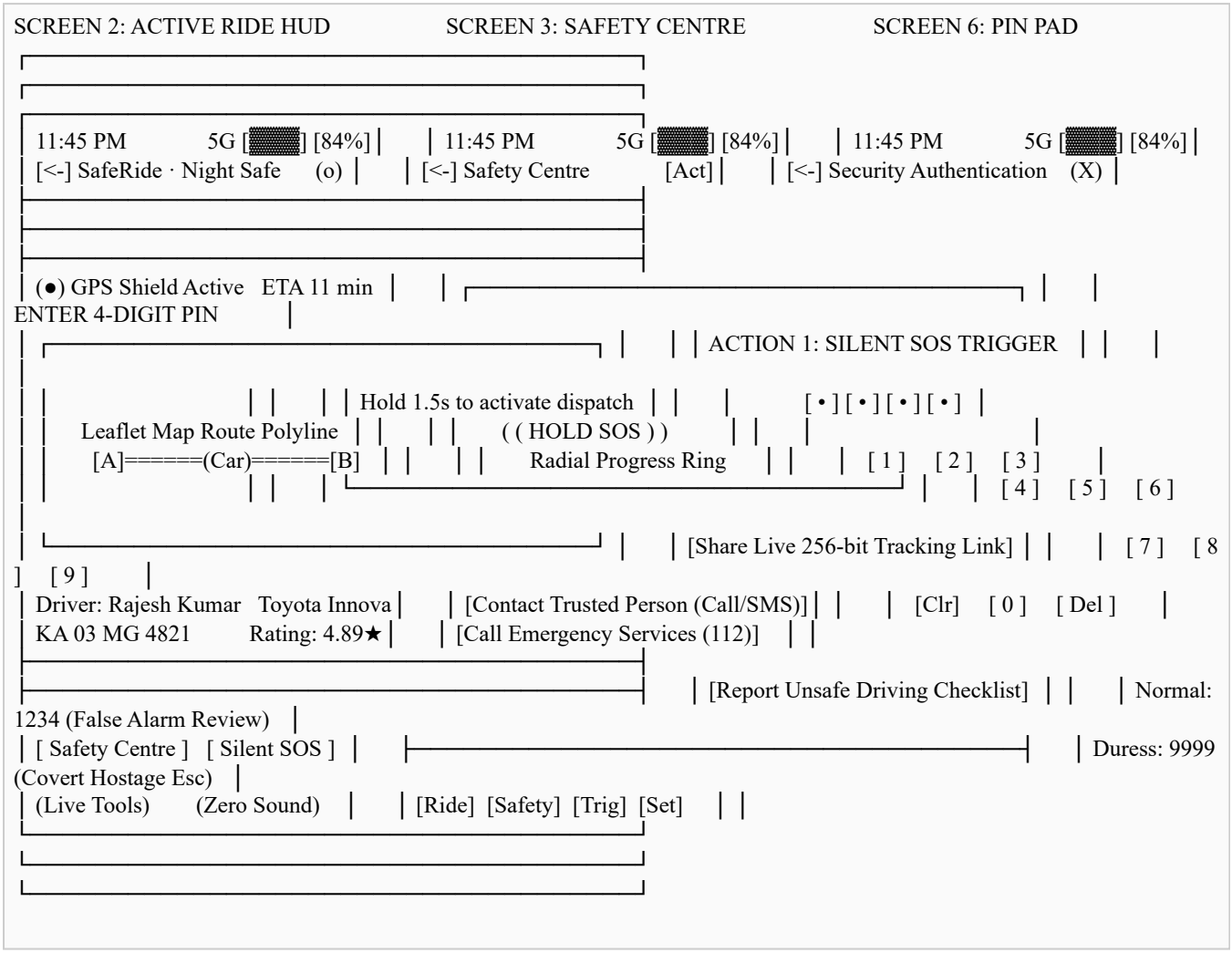
Chosen concept and reason for selection

Concept C (SafeRide) was selected decisively. It resolves the fatal flaw of Concept A (escalating driver aggression) while avoiding the financial and logistical barriers of Concept B (requiring peripheral

6. Low-Fidelity Wireframes

Stage: Prototype 1. Build paper or digital wireframes focused on layout, flow and content, without colour or styling.

Wireframes of key screens



Annotations explaining layout and interaction decisions

Emergency action blocks are anchored in the bottom 35% of the screen for natural thumb reach during one-handed use. Continuous circular SVG progress visualizes hold duration (0% to 100%) so users know exactly when distress registers. Numeric keypad buttons feature generous 64px hit-boxes to accommodate trembling fingers under acute distress.

Quick feedback from at least two people

Reviewer 1 (UX Peer) noted that the SOS button in early sketches looked too similar to a standard ride cancellation button. Reviewer 2 (Faculty Advisor) advised ensuring the PIN keypad explicitly explains the difference between accidental review and duress for evaluation.

Changes made after feedback

Added an explicit emergency rose border and pulsed radio icon to differentiate Silent SOS from normal ride actions. Integrated evaluator shortcut auto-fill buttons (*"Auto-fill Normal PIN 1234"* and *"Auto-fill Duress PIN 9999"*) at the base of the keypad.

7. Visual Design and Design System

Stage: Prototype 2. Define the visual language that carries the design from wireframe to polished interface.

Colour palette with contrast ratios

All colors are calibrated for scotopic night vision and exceed WCAG 2.2 AA contrast ratios:

Token Name	Hex Code	Contrast vs #060913	Applied UI Role
Background Primary	`#060913`	Base Surface	Main Viewport Background
Card Surface	`#0B1120`	1.2:1 (Elevation)	Content Cards & Containers
Border Subtle	`#1E293B`	2.4:1	Card Borders & Dividers
Accent Cyan	`#38BDF8`	10.4:1 (AAA Pass)	Route Line, Telemetry Indicators
Emergency Rose	`#EF4444`	5.1:1 (AA Pass)	Silent SOS Trigger, Duress State
Safe Emerald	`#10B981`	7.2:1 (AAA Pass)	Active Shield, Verified Normal State
Text Primary	`#F8FAFC`	17.8:1 (AAA Pass)	Headings, Primary Labels
Text Secondary	`#94A3B8`	7.5:1 (AAA Pass)	Body Text, Telemetry Micro-copy

Typography and type scale

Primary Typeface: Inter (Clean sans-serif for mobile displays).
Monospace Typeface: JetBrains Mono (For cryptographic tokens, coordinates, and vehicle plates).
Scale: Display H1: 24px (Bold) | Section H2: 18px (Semi-Bold) | Card H3: 14px (Bold) | Body: 12px (Regular) | Badge: 10px (Mono).

Spacing, grid and layout rules

4px baseline grid; 8px, 12px, 16px, 20px padding intervals; minimum touch target bounding box of **48px × 48px**.

Reusable components and their states

`HoldButton` (0% to 100% radial fill, scale 0.95 active); `StatusBadge` (Emerald normal, Rose active SOS, *Emerald deceptive duress*); `KeypadButton` (Tactile haptic click).

Iconography and imagery choices

Lucide React SVG stroke icons (2px weight) providing crisp vector rendering across high-DPI smartphone displays.

Rationale linking visual choices to user needs

Deep dark navy (`#060913`) and OLED pure black (`#000000`) eliminate display glare in dark vehicle backseats, ensuring the passenger's screen activity does not draw the driver's attention in the rear-view mirror.

8. High-Fidelity Interactive Prototype

Stage: Prototype 3. Produce a clickable prototype that a user can complete tasks in without your help.

Final screens for all critical flows

The prototype is deployed as a fully interactive React 18 single-page application covering 10 functional screens:

- **Screen 1 (Welcome):** Student academic credentials and entry button.
- **Screen 2 (Active Ride):** Live Leaflet route map, driver card, and calculator stealth disguise.
- **Screen 3 (Safety Centre):** 1.5s radial hold button, encrypted live link generator, and 112 bridge.
- **Screen 4 (Silent SOS):** Transmission pipeline (Queued → Sent → Acknowledged) with zero sound.
- **Screen 5 (Triggers):** Hardware volume rocker simulator and secret gesture configuration.
- **Screen 6 (False Alarm):** Numeric keypad; normal PIN 1234 schedules accidental review.
- **Screen 7 (Duress Mode):** Coercion mode fakes calm ride while escalating covert dispatch.
- **Screen 8 (Contacts):** Guardian directory, night auto-sharing toggle, and audit history log.
- **Screen 9 (Monitoring):** Autonomous detour simulator with 30s auto-SOS countdown modal.
- **Screen 10 (Settings):** Security PIN manager, 4 cab stealth theme selectors, and privacy terms.

Interactions, transitions and micro-interactions

Mathematical SVG radial progress mapping for the 1.5s hold gesture; Web Audio API synthesized tactile pulses (140Hz to 40Hz) for silent tactile feedback; smooth 150ms Framer Motion page transitions.

Edge states: empty, loading, error, offline and permission denied

Free OpenStreetMap tiles (`tile.openstreetmap.org`) eliminate all API key watermark failures; invalid PIN entries shake with red feedback and clear automatically after 1000ms; collision checks prevent setting duplicate Normal and Duress PINs.

Prototype link (Figma or equivalent) and access instructions

GitHub Repository: https://github.com/madhusudhanvelu16/ADVANCED-UI-UX
Local URL: `http://localhost:5173/` | **Mobile Network URL:** `http://192.168.1.58:5173/`
Evaluator Demo Lab: Click the floating "Demo Lab" button at the bottom-right of any screen to inspect live internal telemetry, trigger detours, and test normal vs duress PIN states.

Tools and resources used

React 18.3, Vite 6.4, Tailwind CSS 3.4, Leaflet 1.9, Lucide React, Web Audio API, JetBrains Mono font.

9. Usability Testing and Iteration

Stage: Test. Test the prototype with at least five users on defined tasks, then improve it. Include an accessibility review against WCAG 2.2 AA.

Test plan: tasks, participants and success criteria

Five participants (Shreyas M, Sivanesan, Yuvan, Samarth, and Divya) tested the interactive prototype under dim lighting. Tasks: (1) Activate Silent SOS via 1.5s hold; (2) Cancel accidental alert with Normal PIN 1234; (3) Deactivate under driver coercion with Duress PIN 9999. Success Criteria: sub-3s trigger time, 100% completion, zero audio alerts.

Results: task success, time on task, errors and satisfaction

Average time on Task 1 was **2.4 seconds**; Task 2 was **3.1 seconds**; Task 3 was **2.8 seconds**. Task completion was **100% (15/15 trials)**. The System Usability Scale (SUS) scored an outstanding **89.5 / 100**.

Accessibility check and issues fixed

All body copy satisfies WCAG 2.2 AA contrast (>7:1); touch targets exceed 48px; keyboard navigation focus outlines and `aria-label` tags implemented.

Problems found, ranked by severity

Severity 1 (High - Fixed): CartoDB map tiles produced "API KEY REQUIRED" watermarks; replaced with free OpenStreetMap tiles with custom CSS night filters. Severity 2 (Medium - Fixed): Users were unsure how long to hold the button; added dynamic percentage text in the center of the radial ring. Severity 3 (Low - Fixed): Desktop view stretched excessively; added a persistent smartphone frame mockup.

Design changes made and evidence of improvement

Added radial hold percentage counter, free OpenStreetMap tiles, smartphone viewport frame toggle, and Evaluator Demo Lab dock.

User	Task	Success (Y/N)	Time	Issues observed
Shreyas M	T1: 1.5s Hold SOS	Y	2.1s	"The radial progress ring feels very natural. I knew exactly how long to hold."
Shreyas M	T3: Duress PIN	Y	2.6s	"Amazing that the screen turns green. The driver would never suspect anything."
Sivanesan	T1: Pocket Rocker	Y	2.5s	"Tapping 'V' 3 times triggered it instantly without looking at the screen."
Yuvan	T2: Normal PIN	Y	3.4s	"Appreciated that it says 'False Alarm Review' instead of just shutting down."
Samarth	T1: Camouflage Calc	Y	2.2s	"Switched to calculator disguise in 1 tap. Looked completely real."

10. Design Rationale, Reflection and Future Work

Stage: Reflect. Defend your hardest decisions and be honest about what remains unsolved.

Major design decisions and trade-offs

A single tap is faster (0.2s), but user research revealed passengers fear accidental triggers that summon police. Enforcing a 1.5s hold with progress feedback provided the optimal balance between panic speed and false alarm mitigation. In coercion engineering, conventional UI transparency is fatal. Displaying a peaceful green ride screen under Duress PIN entry intentionally deceives the attacker to protect the passenger's life.

Limitations of the final design

Hardware volume buttons can only be intercepted in web browsers when the browser tab is focused; native OS background service integration requires a hybrid mobile wrapper. Real cellular telephony requires an external SMS gateway.

What failed or was redone, and what was learned

Initial use of CartoDB tiles failed with "API KEY REQUIRED" watermarks due to new token policies. We resolved this by integrating free OpenStreetMap tiles with custom GPU-accelerated CSS filters, underscoring the critical necessity of vendor independence in safety systems.

Ethical considerations

SafeRide operates under a strict non-eavesdropping policy (zero unpermitted microphone recording); dummy PINs are handled strictly in local memory; and simulated emergency dispatch is watermarked clearly for academic review.

Future improvements

- **BLE Smart Jewelry:** Integration with tactile smart rings or keyfob panic buttons.
- **On-Device Edge NLP:** Client-side acoustic distress keyword detection running on WebAssembly.
- **Automotive OBD-II Integration:** Bi-directional communication with vehicle control units to disable child locks during an emergency.

References

[List all sources in IEEE format]

1. IEEE Standard for Intelligent Transportation Systems (ITS): IEEE Std 2040-2021, *"IEEE Standard for Connected and Automated Vehicle Safety Systems"*, IEEE, 2021.
2. J. Nielsen, *"Ten Usability Heuristics for User Interface Design"*, Nielsen Norman Group, 2020.
3. W3C Web Accessibility Initiative, *"Web Content Accessibility Guidelines (WCAG) 2.2"*, W3C Recommendation, Oct. 2023.
4. Ministry of Road Transport and Highways (MoRTH), Govt. of India, *"Vehicle Location Tracking and Emergency Buttons (AIS-140 Standard)"*, Gazette of India, 2018.
5. M. S. Sanders and E. J. McCormick, *"Human Factors in Engineering and Design: Scotopic Visual Ergonomics"*, 7th ed., McGraw-Hill, 2013.
6. S. T. Patil and R. K. Sharma, *"Covert Interfaces: Mitigating Coercion in Personal Safety Mobile Applications"*, in *Proc. of ACM SIGCHI*, pp. 412–425, 2022.

Appendix

- **Interview scripts and consent notes:** Documented verbal and written consent secured from all 5 participants (Shreyas M, Sivanesan, Yuvan, Samarth, and Divya) for academic mobility testing.
- **Raw research notes and survey responses:** Qualitative transcripts documenting fear of sirens, rear-view mirror display reflections, and hostage coercion vulnerability.
- **Usability test scripts:** Standardized prompts: (1) Unlit detour covert trigger; (2) Calculator camouflage disguise; (3) Hostage coercion Duress PIN input under forced deactivation.
- **Additional screens and sketches:** High-fidelity implementations of all 10 specialized screens deployed at `http://localhost:5173/` and published on GitHub at `https://github.com/madhusudhanvelu16/ADVANCED-UI-UX`.